

13. A helical gear with 20° pressure angle and 30° helix angle mounted at the midspan of a shaft that is supported between two bearings at the ends. The nature of the stresses induced in the shaft is

(GATE-20-SET-1)

- (a) normal stress due to bending in one plane and axial loading ; shear stress due to torsion.
- (b) normal stress due to bending in two planes and axial loading ; shear stress due to torsion.
- (c) normal stress due to bending in two planes; shear stress due to torsion.
- (d) normal stress due to bending only

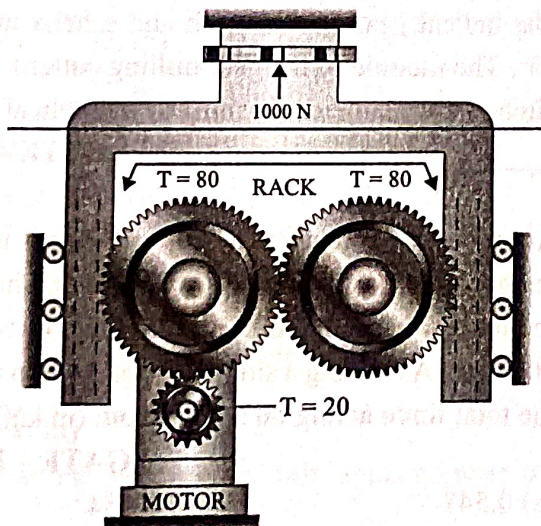
Two Marks Questions

01. In order to test the efficiency of reducer gear train 1kW input was given at the input end at a speed of 1440 rpm and at the output end the measured torque was 56.36 N.m. If the ratio of speed reduction in this unit is 10:1, the efficiency is about

(GATE-ME-89)

- (a) 78 % (b) 85% (c) 63% (d) 96%

Common Data Questions Q.Nos. 02 and 03.



A compacting machine shown in the figure below is used to create a desired thrust force by using a rack and pinion arrangement. The input gear is mounted on the motor shaft. The gears have involute teeth of 2mm module.

(GATE-ME-04) (2M)

02. If the drive efficiency is 80%, the torque required on the input shaft to create 1000N output thrust is
- (a) 20 Nm (b) 25 Nm
 - (c) 32 Nm (d) 50 Nm

03. If the pressure angle of the rack is 20° , the force acting along the line of action between the rack and the gear teeth is

- (a) 250 N (b) 342 N
- (c) 532 N (d) 600 N

04. Twenty degree full depth involute profiled 19-tooth pinion and 37-tooth gear are in mesh. If the module is 5 mm, the center distance between the gear pair will be

(GATE-ME-06)

- (a) 140 mm (b) 150 mm
- (c) 280 mm (d) 300 mm

Statement for Linked Answer to Q. Nos. 05 to 07

A gear set has a pinion with 20 teeth and a gear with 40 teeth. The pinion runs at 30 rev/s and transmits a power of 20 KW. The teeth are on the 20° full-depth system and have a module of 5mm. The length of the line of action is 19 mm.

(GATE-ME-07)

05. The center distance for the above gear set in mm is
- (a) 140 (b) 150 (c) 160 (d) 170

06. The contact ratio of the contacting tooth is
- (a) 1.21 (b) 1.25 (c) 1.29 (d) 1.33

07. The resultant force on the contacting gear tooth is N is
- (a) 77.23 (b) 212.20
 - (c) 2258.1 (d) 289.43

08. A spur gear has a module of 3 mm, number of teeth 16, a face width of 36 mm and a pressure angle of 20° . It is transmitting a power of 3 KW at 20 rev/s. Taking a velocity factor of 1.5 and a form factor of 0.3, the stress in the gear tooth is about

(GATE-ME-08)

- (a) 32MPa (b) 46MPa
(c) 58MPa (d) 70MPa

09. Match the type of gears with their most appropriate description (GATE-ME-08)

Type of gear

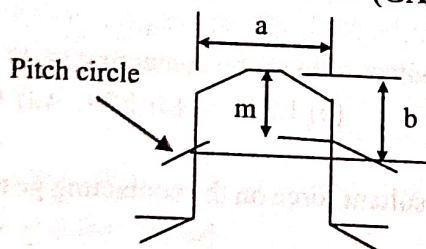
- P. Helical
Q. Spiral Bevel
R. Hypoid
S. Rack and pinion

Description

1. Axes non parallel and non intersecting
2. Axes parallel and teeth are inclined to the axis
3. Axes parallel and teeth are parallel to the axis
4. Axes are perpendicular and intersecting and teeth are inclined to the axis
5. Axes are perpendicular and used for large speed reduction
6. Axes parallel and one of the gears has infinite radius

- (a) P-2,Q-4,R-1,S-6 (b) P-1,Q-4,R-5,S-6
(c) P-2,Q-6,R-4,S-2 (d) P-6,Q-3,R-1,S-5

10. One tooth of gear having 4 module and 32 teeth is shown in the figure. Assume that the gear tooth and the corresponding tooth space make equal intercepts on the pitch circumference. (GATE-ME-08)



The dimensions 'a' and 'b' respectively, are closest to

- (a) 6.08 mm, 4 mm (b) 6.48 mm, 4.2 mm
(c) 6.28 mm, 4.3 mm (d) 6.28 mm, 4.1 mm

Statement for Linked Answer Q.Nos. 11 & 12

A 20° full depth involute spur pinion of 4 mm module and 21 teeth is to transmit 15 kW at 960 rpm. Its face width is 25 mm. (GATE-ME-09)

11. The tangential force transmitted (in N) is
(a) 3552 (b) 2611 (c) 1776 (d) 1305

12. Given that the tooth geometry factor is 0.32 and the combined effect of dynamic load and allied factors intensifying the stress is 1.5; the minimum allowable stress (in MPa) for the gear material is
(a) 242.0 (b) 166.5
(c) 121.0 (d) 74.0

13. A pair of spur gears with module 5 mm and a center distance of 450 mm is used for a speed reduction of 5:1. The number of teeth on pinion is _____ (GATE-ME-14)

14. A spur pinion of pitch diameter 50 mm rotates at 200 rad/s and transmits 3 kW power. The pressure angle of the tooth of the pinion is 20° . Assuming that only one pair of the teeth is in contact, the total force (in Newton) exerted by a tooth of the pinion on the tooth on a mating gear is _____ (GATE-ME-14-SET-2)

15. A helical gear with involute tooth profile has been machined with a disc-type form gear milling cutter. The helical gear has 30 teeth and a helix angle of 30° . The module of the gear milling cutter is 2. The pitch circle diameter (in mm) of the helical gear is _____ (GATE - PI-16)

16. A pair of spur gears with 20° full-depth involute teeth is used to transmit 3.5 kW of power. The pinion rotates at 700 rpm and has pitch circle diameter of 100 mm. Assuming a single pair of teeth in contact, the total force acting on a gear tooth (in kN) is _____ (GATE - PI - 17)
(a) 0.347 (b) 0.954
(c) 1.016 (d) 1.302

17. A steel spur pinion has a module (m) of 1.25 mm, 20 teeth and 20° Pressure angle. The pinion rotates at 1200 rpm and transmits power to a 60 teeth

gear. The face width (F) is 50 mm, Lewis form factor $Y = 0.322$ and a dynamic factor $K_v = 1.26$. The bending stress (σ) induced in a tooth can be calculated by using the Lewis formula given below. If the maximum bending stress experienced by the pinion is 400 MPa, the power transmitted is _____ kW (round off to one decimal place).

$$\text{Lewis formula: } \sigma = \frac{K_v W^t}{F m Y},$$

where W^t is the tangential load acting on the pinion.

(GATE-20-SET-2)

18. A pair of spur gears is designed to transmit 20 kW power at a pitch line velocity of 10 m/s. Diameter of the driving gear is 0.5 m. The tangential force, in N, between the driver and the driven gear is _____ (in integer). (GATE-PI-23)

KEY & Detailed Solutions

ONE MARK QUESTIONS

01. (a)	02. (d)	03. (*)	04. True	05. (*)
06. (c)	07. (c)	08. (d)	09. (b)	10. (c)
11. (d)	12. (d)	13. (b)		

TWO MARKS QUESTIONS

01. (b)	02. (b)	03. (c)	04. (a)	05. (b)
06. (c)	07. (c)	08. (b)	09. (a)	10. (d)
11. (a)	12. (b)	13. 30	14. 638.5 N	
15. 69.28	16. (c)	17. 10	18. 2000	

One Mark Solutions

01. Ans: (a)

Sol: To avoid interference, minimum number of tooth

$$\text{required } T_{\min} = \frac{2a_r}{\sin^2 \phi}$$

where, ϕ = pressure angle

$a_r = 1$ for full depth gear

02. Ans: (d)

Sol: In case of involute gearing, centre distance does not effect law of gearing, pressure angle and hence velocity ratio.

03. Ans: Decreases, Increases

Sol: Larger pressure angle creates smaller base circle. In this case, the tip of the tooth of one gear will not have a chance to contact the flank of the other gear which reduces chances of interference and hence, undercutting decreases.

04. Ans: True

Sol: If the addendum circles of both the gears intersect common tangent to the base circles within the points of tangency, then it will not intersect the base circle i.e. no interference is occurred.

05. Ans: A - 4, B - 3, C - 2, D - 1

Sol:

- Worm gears have high speed reduction ratio upto 100:1.
- Cross helical gear sets are used to transmit power and motion between non-intersecting and non parallel axes.
- Bevel gears are used to connect two shafts axes of which are intersecting and coplanar.
- Spur gears are used to transmit power when the shafts are parallel.

06. Ans: (c)

Sol: Pitch circle is an imaginary circle which by pure rolling action, would give the same motion as the actual gear.

- An involute of a circle is a plane curve generated by a point on a tangent, which rolls on the circle without slipping or by a point on a taut string which is unwrapped from a reel. In connection with toothed wheels, the circle is known as base circle.

- Addendum circle is the circle drawn through the top of the teeth and is concentric with the pitch circle.

07. Ans: (c)

Sol: Let, μ = coefficient of friction, γ = lead angle,

$$l = \text{lead} = n \times p,$$

(here, n = number starts, p = pitch)

d = pitch circle diameter of worm,

$$\tan \gamma = \frac{l}{\pi d}$$

- Self-locking condition, $\mu > \tan \gamma$.
- If $\mu < \tan \gamma$ i.e friction is not sufficient to hold, the drive will be reversible.
- If number of starts are increased, lead will increase, which will eventually increase lead angle. And if $\tan \gamma$ becomes more than ' μ ', the drive will be reversible.

08. Ans: (d)

Sol: Velocity ratio = $\frac{1440}{36} = 40$

09. Ans: (b)

Sol:

Gear Type	Velocity ratio
Spur or Helical Gears	Maximum 10 : 1 Normally 06 : 1
Bevel gears	Maximum 3 : 1 Normally 1 : 1
Worm gears	Maximum 100 : 1

10. Ans: (c)

Sol: $T_{\min} = \frac{2}{\sin^2 \phi} = \frac{2}{\sin^2 20^\circ} = 18$

11. Ans: (d)

Sol: $T_p = 40$, $T_G = 120$, $N_p = 1200$ rpm

Torque by pinion = 20 N-m.

Torque by gear = $20 \times \frac{T_o}{T_p} = 20 \times \frac{120}{40} = 60$ N-m.

12. Ans: (d)

Sol: Interference can be avoided by using more number of teeth. If number of teeth is increased then size of the gear will be increased which is desirable to avoid interference.

- Larger pressure angle creates smaller base circle, so that a greater portion of the tooth profile has an involute shape. In this case, the tip of the tooth of one gear will not have a chance to contact the flank of the other gear on its non-involute portion.

13. Ans: (b)

Sol: A helical gear mounted at the midspan of a shaft induces -

- Normal stress due to bending in two planes and axial loading
- Shear stress due to torsion

Two Marks Solutions

01. Ans: (b)

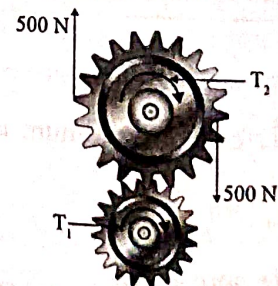
Sol: Out-put shaft speed = $1440 \times \frac{1}{40} = 144$ rpm

Efficiency, $\eta = \frac{\text{output power}}{\text{input power}}$

$$= \frac{T_o \omega_o}{1 \text{ kW}} = \frac{56.36 \times \frac{2\pi \times 144}{60}}{1 \times 10^3 \text{ Watts}} = 84.9\%$$

02. Ans: (b)

Sol:



Thrust shared by each gear = 500 N
Torque required

$$T_2 = \frac{1000 \times m z}{\eta \times 2} = \frac{500 \times 2 \times 80}{1000 \times 0.8} = 100 \text{ N-m}$$

$$T_2 \omega_2 = T_1 \omega_1$$

$$T_1 = T_2 \times \frac{\omega_2}{\omega_1} = T_2 \times \frac{z_1}{z_2} = 100 \times \frac{20}{80}$$

$$T_1 = T_{\text{input}} = 25 \text{ N-m}$$

03. Ans: (c)

Sol: $F \cos \phi = 500$

$$F = \frac{500}{\cos \phi} = 532 \text{ N}$$

04. Ans: (a)

Sol: $C = \frac{m(T_P + T_G)}{2} = \frac{5(19 + 37)}{2} = 140 \text{ mm}$

05. Ans: (b)

Sol: $C = \frac{m(T_P + T_G)}{2} = 5 \times \frac{(20 + 40)}{2} = 150 \text{ mm}$

06. Ans: (c)

Sol: Contact ratio,

$$m_c = \frac{\text{Length of action}}{P_b} = \frac{L_{ab}}{p_c \cos \phi}$$

$$= \frac{19 \text{ mm}}{\pi \times 5 \times \cos 20} = 1.287$$

07. Ans: (c)

Sol: $F_t = \frac{\text{Power}}{V} = \frac{20 \times 10^3}{\frac{\pi \times 5 \times 20 \times 30}{1000}} = 20.123 \text{ kN}$

$$F = \frac{F_t}{\cos \phi} = \frac{2.123}{\cos 20^\circ} = 2.259 \text{ kN}$$

08. Ans: (b)

Sol: $F_t = S w m y C_v$

$$= \frac{3 \times 10^3 \text{ Watt}}{\frac{\pi \times 3 \times 16 \times 20}{1000}} = S \times 36 \times 3 \times 0.3 \times \frac{1}{1.5}$$

$$S = 46.04 \text{ MPa}$$

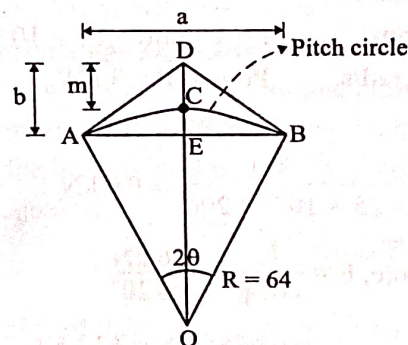
09. Ans: (a)

Sol:

- In helical gear the teeth are inclined to the axis and they are used for parallel and co-planar shafts.
- Spiral bevel gears are mounted on shaft whose axes are intersecting.
- Hypoid bevel gears are used to connect shafts whose axes do not intersect. They are very similar to spiral gears. However, their pitch surfaces are hyperboloids rather than cones.
- If the gear of a shaft meshes externally and internally with the gear wheel of infinite radius, it is called rack and pinion.

10. Ans: (d)

Sol:



Angle made by 32 teeth + 32 tooth space = 360° .

$$2\theta = \frac{360}{64} = 5.625$$

$$\theta = 2.8125$$

$$R = \frac{mT}{2} = \frac{4 \times 32}{2} = 64 \text{ mm}$$

$$a = R \sin \theta \times 2 = 64 \times \sin (2.81) \times 2 = 6.28 \text{ mm}$$

$$OE = R \cos \theta = 64 \times \cos (2.8125) = 63.9 \text{ mm}$$

$$b = \text{addendum} + CE = \text{module} + (OC - OE)$$

$$= 4 + (64 - 63.9) = 4.1 \text{ mm}$$

11. Ans: (a)

Sol: $F_t = \frac{P}{V} = \frac{15 \times 10^3 \text{ W}}{\frac{\pi \times 4 \times 21}{1000} \times \frac{960}{60}} = 3552 \text{ N}$

12. Ans: (b)

$$\text{Sol: } F_t = \frac{\text{Swamy} \times C_v}{\text{Stress intensity factor}}$$

$$3552 = \frac{S \times 25 \times 4 \times 0.32 \times 1}{1.5}$$

$$S = 166.5$$

13. Ans: 30

$$\text{Sol: } m = 5 \text{ mm, } C = 450 \text{ mm}$$

$$\frac{T_G}{T_P} = \frac{N_P}{N_G} = 5, \Rightarrow T_G = 5T_P$$

$$C = \frac{m(T_P + T_G)}{2} \Rightarrow 450 = \frac{5(T_P + 5T_P)}{2}$$

$$\Rightarrow T_P = 30$$

14. Ans: 638.5 N

$$\text{Sol: } d = 50 \text{ mm, } r = 25 \text{ mm} = 25 \times 10^{-3} \text{ m,}$$

$$\omega = 200 \text{ rad/s, } \text{Power, } P = 3 \text{ kW}$$

$$\phi = 20^\circ, \quad V = r\omega$$

$$F_t = \frac{P}{V} = \frac{3}{25 \times 10^{-3} \times 200} = 0.6 \text{ kN}$$

$$\text{Total force, } F = \frac{F_t}{\cos \phi} = \frac{0.6 \text{ kN}}{\cos 20^\circ} = 0.6385 \text{ kN} = 638.5 \text{ N}$$

15. Ans: (69.28)

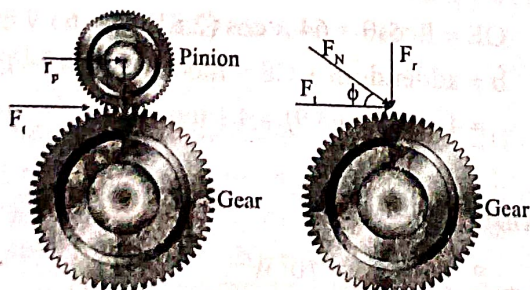
$$\text{Sol: } m_n = \text{normal module} = 2 \text{ mm}$$

Pitch circle diameter,

$$d = \frac{Zm_n}{\cos \psi} = \frac{30 \times 2}{\cos 30} = 69.28 \text{ mm}$$

16. Ans: (c)

Sol:



$$\phi = 20^\circ, \quad \text{Power} = 3.5 \text{ kW}$$

$$N_{\text{pinion}} = 700 \text{ rpm, } r_{\text{pinion}} = 50 \text{ mm} = 0.05 \text{ m}$$

$$\text{Power} = \frac{2\pi \times N_p \times F_t \times r_p}{60}$$

$$3500 = \frac{2\pi \times 700 \times F_t \times 0.05}{60}$$

$$F_t = 954.92 \text{ N}$$

Resultant force,

$$F_N = \frac{F_t}{\cos \phi} = \frac{954.92}{\cos 20} = 1016.21 \text{ N}$$

$$F_N = 1.016 \text{ kN}$$

17. Ans: 10

$$\text{Sol: Given: } m = 1.25 \text{ mm, } z = 20$$

$$\text{Diameter of pinion} = M.z = 25 \text{ mm} = 0.025 \text{ m}$$

Power transmitted by pinion

$$P = \frac{2\pi N.T}{60}$$

$$= \frac{2\pi N.W^t \cdot \frac{d_p}{2}}{60}$$

$$\text{But } \sigma = \frac{K_v.W^t}{F.m.y}$$

$$W^t = \frac{\sigma.F.m.Y}{K_v}$$

$$\therefore \text{Power} = \frac{2\pi N \left(\frac{\sigma.F.m.Y}{K_v} \right) \frac{d_p}{2}}{60}$$

$$= \frac{2 \times p \times 1200 \times 400 \times 50 \times 1.25 \times 0.322}{1.26} \times \frac{0.025}{2}$$

$$P = 10035.64 \text{ W}$$

$$P = 10.035 \text{ kW}$$

$$P \approx 10 \text{ kW}$$

18. Ans: 2000

$$\text{Sol: Power transmitted, } P = F_t \times V$$

where F_t = tangential load

V = tangential velocity

$$20 \times 10^3 = F_t \times 10$$

$$\Rightarrow F_t = 2000 \text{ N}$$

Chapter **5** Rolling Contact Bearings

One Mark Questions

01. The expected life of a ball bearing subjected to a load of 9800 N and working at 1000 RPM is 3000 hours. What is the expected life of the same bearing for a similar load of 4900 N and a speed of 2000 rpm
(GATE-ME-87)
(a) Unchanged (b) 12,000 hours
(c) 1,500 hours (d) 6,000 hours
02. If the load on a ball bearing is reduced to half, the life of the ball bearing will (GATE-ME-88)
(a) increase 8 times (b) increase 4 times
(c) increase 2 times (d) not change
03. Match the rolling element bearings with the most appropriate loading condition (GATE-ME-91)
- | | |
|--|--|
| Bearing - type | |
| a. Ball bearing | |
| b. Roller bearing | |
| c. Needle bearing | |
| d. Taper roller bearing | |
| Loading condition | |
| p. Tangential load | |
| q. Radial load | |
| r. Heavy radial load with impact | |
| s. Light radial load with space limitation | |
| (t) Heavy radial and axial load | |
| (u) Fatigue load | |
04. Spherical roller bearings are normally used (GATE-ME-92)
(a) for increased radial load
(b) for increased thrust load
(c) when there is less radial load
(d) to compensate for angular misalignment

05. The basic load rating of a ball bearing is (GATE-ME-98)
(a) the maximum static radial load that can be applied without causing any plastic deformation of bearing components
(b) the radial load at which 90% of the group of apparently identical bearings run for one million revolutions before the first evidence of failure
(c) the maximum radial load that can be applied during operation without any plastic deformation of bearing components
(d) a combination of radial and axial loads that can be applied without any plastic deformation
06. Ball bearings are rated by a manufacturer for a life of 10^6 revolutions. The catalogue rating of a particular bearing is 16 kN. If the design load is 2 kN, the life of the bearing will be $P \times 10^6$ revolutions, where P is equal to (GATE-ME-14-SET-4)
07. Which of the bearings given below SHOULD NOT be subjected to a thrust load? (GATE - 16 - SET - 3)
(a) Deep groove ball bearing
(b) Angular contact ball bearing
(c) Cylindrical (straight) roller bearing
(d) Single row tapered roller bearing

Two Marks Questions

01. The dynamic load capacity of 6306 bearing is 22 kN. The maximum radial load it can sustain to operate at 600 rev/min, for 2000 hours is (GATE-ME-97)
(a) 4.16 kN (b) 3.60 kN
(c) 6.25 kN (d) 5.29 kN

02. Match the following (GATE-ME-97)

List - I

- Automobile wheel mounting on axle
- High speed grinding spindle
- IC-Engine connecting rod
- Leaf spring eye mounting

List - II

- Magneto bearing
- Angular contact bearing
- Taper roller bearing
- Hydrodynamic journal bearing
- Sintered metal bearing
- Teflon/Nylon bush

03. The life of a ball bearing at a load of 10 kN is 8000 hours. Its life in hours, if the load is increased to 20 kN, keeping all other conditions same, is

(GATE-ME-00)

- | | |
|----------|----------|
| (a) 4000 | (b) 2000 |
| (c) 1000 | (d) 500 |

 04. A ball bearing operating at a load F has 8000 hours of life. The life of the bearing, in hours, when the load is doubled to $2F$ is

(GATE-ME-07)

- | | |
|----------|----------|
| (a) 8000 | (b) 6000 |
| (c) 4000 | (d) 1000 |

05. Two identical ball bearings P and Q are operating at loads 30 kN and 45 kN respectively. The ratio of the life of bearing P to the life of bearing Q is

(GATE-ME-11)

- | | |
|-----------|----------|
| (a) 81/16 | (b) 27/8 |
| (c) 9/4 | (d) 3/2 |

 06. For ball bearings, the fatigue life L measured in number of revolutions and the radial load F are related by $FL^{1/3} = K$, where K is a constant. It withstands a radial load of 2 kN for a life of 540 million revolutions. The load (in kN) for a life of one million revolutions is _____

(GATE -15 -Set 3)

 07. A self-aligning ball bearing has a basic dynamic load rating (C_{10} for 10^6 revolutions) of 35 kN. If the equivalent radial load on the bearing is 45 kN, the expected life (in 10^6 revolutions) is

(GATE - 18 - SET - 1)

- | | |
|----------------|----------------|
| (a) below 0.5 | (b) 0.5 to 0.8 |
| (c) 0.8 to 1.0 | (d) above 1.0 |

KEY & Detailed Solutions

ONE MARK QUESTIONS

01. (b)	02. (a)	03. (*)	04. (d)	05. (b)
06. 512 N	07. (c)			

TWO MARKS QUESTIONS

01. (d)	02. (*)	03. (c)	04. (d)	05. (b)
06. 15 to 17	07. (a)			

One Mark Solutions

01. Ans: (b)

$$\text{Sol. } \frac{L_1}{L_2} = \left(\frac{P_2}{P_1} \right)^3$$

$$\Rightarrow \frac{60 \times 1000 \times 3000}{10^6} = \left(\frac{4900}{9800} \right)^3$$

$$\Rightarrow L_h = 1500 \times 8 = 12,000 \text{ hrs}$$

02. Ans: (a)

$$\text{Sol: For ball bearing, life} \propto \left(\frac{1}{\text{Equivalent radial load}} \right)^3$$

03. Ans: a - u, b - q, c - r, d - t.

Sol:

- Ball bearings are suitable for cyclic loads and speeds.
- Roller bearings can take only radial load and they can't take thrust load.

Needle bearings are like cylindrical bearings with small diameter and relatively large length. They are best suited for applications involving oscillatory motion such as piston pin bearings.

In taper bearings, the reaction from race is inclined to shaft axis. It allows bearing to take both radial and axial loads. It can take heavy loads.

04. Ans: (d)

Sol: Spherical roller bearings support a rotating shaft in the bore of the inner ring that may be misaligned with respect to the outer ring.

05. Ans: (b)

Sol: The rating life of a group of apparently identical ball bearings is defined as the number of revolutions that 90% of the bearings will complete or exceed before the first evidence of fatigue crack.

06. Ans: 512 N

Sol: $C = 16 \text{ kN}$, $P = 2 \text{ kN}$

$$L = \left(\frac{C}{P}\right)^n, n = 3 \text{ for ball bearing}$$

$$L = \left(\frac{16}{2}\right)^3 = 512 \text{ million revolutions}$$

$$= 512 \times 10^6 \text{ revolutions} = P \times 10^6$$

$$\therefore P = 512 \text{ N}$$

07. Ans: (c)

Sol: Cylindrical roller bearing can carry only radial loads. It can not take thrust load.

Two Marks Solutions

01. Ans: (d)

Sol:

$$L = \frac{60nL_H}{10^6} = \frac{60 \times 600 \times 2000}{10^6}$$

$$= 72 \text{ million rev}$$

$$L = \left(\frac{C}{P}\right)^3 \Rightarrow 72 = \frac{22^3}{p^3} \Rightarrow P = 5.29 \text{ kN}$$

02. Ans: a-3, b-2, c-4, d-6,

Sol:

- Wheel axle is subjected to heavy axial and radial loads hence taper roller bearing should be used.
- Angular contact bearings provide high thrust capacity which ensures rigidity of axis. Hence they are used in high speed grinding spindle.
- At high load and high speed, rolling bearings become noisy. Also load carrying capacity of rolling bearings decrease at high speeds. Hence, for connecting rod, hydrodynamic journal bearings are used.

03. Ans: (c)

$$\text{Sol: } \frac{L_1}{L_2} = \left(\frac{P_2}{P_1}\right)^3 \Rightarrow \frac{8000 \text{ hrs}}{L_2 \text{ hrs}} = \left(\frac{20}{10}\right)^3 \Rightarrow L_2 = 1000 \text{ hrs}$$

04. Ans: (d)

$$\text{Sol: } \frac{L_1}{L_2} = \left(\frac{P_2}{P_1}\right)^3 \Rightarrow \frac{8000}{L_2} = \left(\frac{2}{1}\right)^3 \Rightarrow L_2 = 1000 \text{ hrs}$$

05. Ans: (b)

$$\text{Sol: } \frac{L_P}{L_Q} = \left(\frac{W_Q}{Q_P}\right)^3 = \left(\frac{45}{30}\right)^3 = \left(\frac{9}{6}\right)^3 = \frac{27}{8}$$

06. Ans: 15 to 17

$$\text{Sol: } FL^{1/3} = k \Rightarrow F_1 L_1^{1/3} = F_2 L_2^{1/3}$$

$$2 \times (540)^{1/3} = F \times (1)^{1/3}$$

$$F = 16.286 \text{ kN}$$

07. Ans: (a)

Sol: For ball bearings

$$L = \left(\frac{C}{P}\right)^3 \text{ in million revolutions}$$

$$\therefore L = \left(\frac{35}{45}\right)^3 = 0.47 \text{ million rev}$$

Chapter

6

Sliding Contact Bearings

One Mark Questions

01. Starting friction is low in (GATE-ME-92)
 - (a) hydrostatic lubrication
 - (b) hydrodynamic lubrication
 - (c) mixed (or semi-fluid) lubrication
 - (d) boundary lubrication
02. In thick film hydrodynamic journal bearings, the coefficient of friction (GATE-ME-96)
 - (a) increases with increase in load
 - (b) is independent of load
 - (c) decreases with increase in load
 - (d) may increase or decrease with increase in load
03. To restore stable operating condition in a hydrodynamic journal bearing, when it encounters higher magnitude loads (GATE-ME-97)
 - (a) oil viscosity is to be decreased
 - (b) oil viscosity is to be increased
 - (c) oil viscosity index is to be increased
 - (d) oil viscosity index is to be decreased
04. Which one of the following is criterion in the design of hydrodynamic journal bearings? (GATE-ME-05)
 - (a) Sommerfeld number
 - (b) Rating life
 - (c) Specific dynamic capacity
 - (d) Rotation factor

Two Marks Questions

01. A natural feed journal bearing of diameter 50 mm and length 50 mm operating at 20 revolutions/sec carries a load of 2 kN. The lubricant used has a viscosity of 20 mPa-s. The radial clearance is 50 μ m. The Sommerfeld number for the bearings is (GATE-ME-07)
 - (a) 0.062
 - (b) 0.125
 - (c) 0.250
 - (d) 0.785
02. A journal bearing has a shaft diameter of 40 mm and a length of 40 mm. The shaft is rotating at 20 rad/s and viscosity of the lubricant is 20 mPa.s. The clearance is 0.020 mm. The loss of the torque due to viscosity of the lubricant is approximately (GATE-ME-08)
 - (a) 0.040 Nm
 - (b) 0.252 Nm
 - (c) 0.400 Nm
 - (d) 0.652 Nm
03. A hydrodynamic journal bearing is subject to 2000 N load at a rotational speed of 2000 rpm. Both bearing bore diameter and length are 40 mm. If radial clearance is 20 μ m and bearing is lubricated with an oil having viscosity 0.03 Pa.s, the Sommerfeld number of the bearing is (GATE-14-SET-1)

KEY & Detailed Solutions

ONE MARK QUESTIONS

01. (a)	02. (c)	03. (b)	04. (a)
---------	---------	---------	---------

TWO MARKS QUESTIONS

01. (b)	02. (a)	03. 0.8
---------	---------	---------

One Mark Solutions

01. Ans: (a)

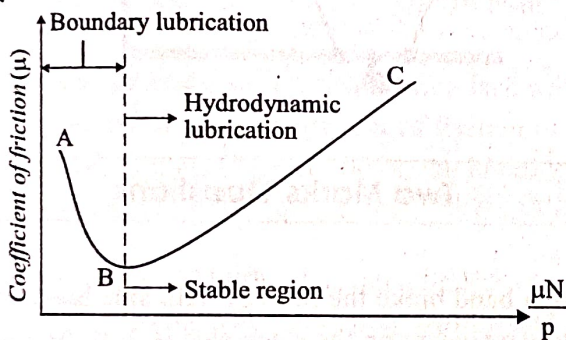
Sol: Hydrodynamic bearings exhibit high starting friction due to metal to metal contact during starting up. Hydrostatic bearings exhibit low starting friction compare to hydrodynamic bearings.

02. Ans: (c)

Sol: $f \propto \frac{\mu N}{p} \propto \frac{\mu N \times \ell d}{W} \left(\because p = \frac{W}{\ell d} \right)$

03. Ans: (b)

Sol:



If hydrodynamic journal bearing encounters higher magnitude loads, $\frac{\mu N}{p}$ decreases, which leads the bearing to be operated in boundary lubrication which is unstable. To avoid this, oil viscosity (μ) is to be increased to counter the increment in load.

04. Ans: (a)

Sol: Sommerfeld number is the criterion in the design of hydrodynamic journal bearings. It contains all the variables which are controlled by the designer.

Sommerfeld Number = $\left(\frac{\mu N_s}{p} \right) \left(\frac{r}{C} \right)^2$

Two Marks Solutions

01. Ans: (b)

Sol: $S = \left(\frac{\mu N_s}{p} \right) \left(\frac{r}{C} \right)^2$

$$= \left(\frac{20 \times 10^{-3} \times 20}{2 \times 10^3} \right) \left(\frac{25}{50 \times 10^{-3}} \right)^2$$

$$= \frac{1}{8} = 0.125$$

02. Ans: (a)

Sol: Torque = $F \times r = \tau \times A \times r$

$$= \frac{\mu V}{C} \times 2\pi r \ell \times r = \mu \times \frac{2\pi r N_s}{C} \times 2\pi r \ell \times r$$

$$= \mu \times \frac{r \omega}{C} \times 2\pi r \ell \times r$$

$$= 20 \times 10^{-3} \times \frac{0.02 \times 20 \times 2\pi}{0.020 \times 10^{-3}} \times 2\pi \times 0.02 \times 0.04 \times 0.02$$

$$= 0.040 \text{ N-m}$$

03. Ans: 0.8

Sol: $W = 2000 \text{ N}$,

$N = 2000 \text{ rpm}$,

$l = d = 40 \text{ mm}$,

$C = 20 \times 10^{-6} \text{ m}$,

$\mu = 0.03 \text{ Pa-s}$

Pressure, 'P' = $\frac{\text{Radial load (W)}}{\text{projected area}}$

$$= \frac{W}{\ell \times d} = \frac{2000}{0.04 \times 0.04} = 125 \times 10^4 \text{ Pa}$$

Sommerfeld number, $S = \left(\frac{\mu N_s}{p} \right) \left(\frac{r}{C} \right)^2$

$$= \left(\frac{0.03 \times \left(\frac{2000}{60} \right)}{125 \times 10^4} \right) \left(\frac{20 \times 10^{-3}}{20 \times 10^{-6}} \right)^2 = 0.8$$

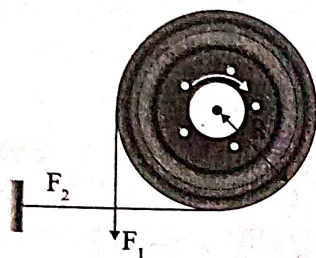
Chapter

7

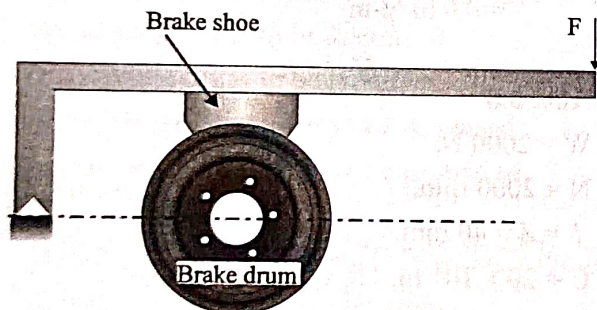
Brakes

One Mark Questions

01. The force F_1 and F_2 in a brake band and the direction of rotation of the drum are as shown in the figure. The coefficient of friction is 0.25. The angle of wrap is $3\pi/2$ radians. It is given that $R = 1$ m and $F_2 = 1$ N. The torque (in N-m) exerted on the drum is _____. (GATE - 16 - SET - 2)

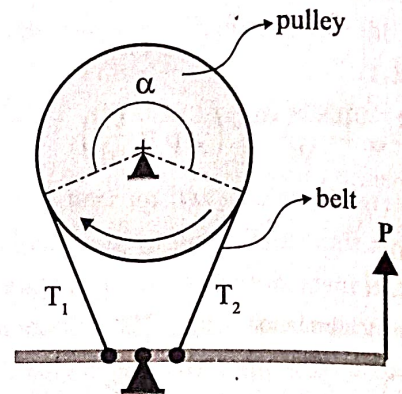


02. For the brake shown in the figure, which one of the following is TRUE? (GATE - 16 - SET - 2)



- (a) Self energizing for clockwise rotation of the drum
(b) Self energizing for anti-clockwise rotation of the drum
(c) Self energizing for rotation in either direction of the drum
(d) Not of the self energizing type
03. The braking system shown in the figure uses a belt to slow down a pulley rotating in the clockwise direction by the application of a force P . The belt wraps around the pulley over an angle $\alpha = 270$

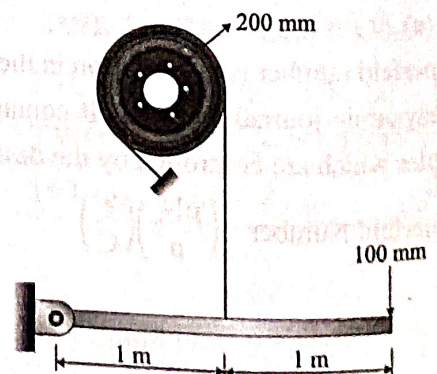
degrees. The coefficient of friction between the belt and the pulley is 0.3. The influence of centrifugal forces on the belt is negligible. During braking, the ratio of the tensions T_1 to T_2 in the belt is equal to _____. (Rounded off to two decimal places) Take $\pi = 3.14$. (GATE_ME-23)



Two Marks Questions

01. In a band brake the ratio of tight side band tension to the tension on the slack side is 3. If the angle of overlap of band on the drum is 180° the coefficient of friction required between drum and the band is _____. (GATE-ME-03)
- (a) 0.20 (b) 0.25 (c) 0.30 (d) 0.35

Statement for Linked Answer Q.Nos. 02 & 03



A band brake consists of a lever attached to one end of the band. The other end of band is fixed to the ground. The wheel has a radius of 200 mm and the wrap angle of band is 270° . The braking force applied to the lever is limited to 100 N and the coefficient of friction between the band and the wheel is 0.5. No other information is given. (GATE-ME-05)

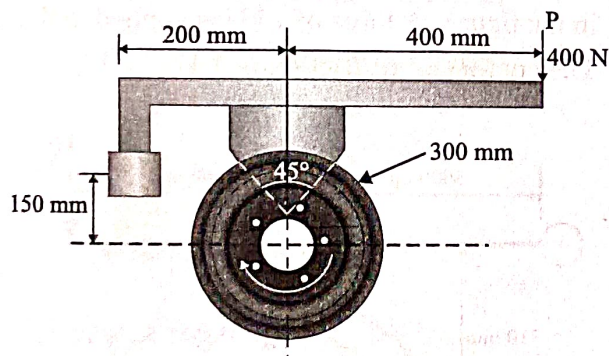
02. The maximum tension that can be generated in the band during braking is

- (a) 1200 N (b) 2110 N
(c) 3224 N (d) 4420 N

03. The maximum wheel torque that can be completely braked is

- (a) 200 N-m (b) 382 N-m
(c) 604 N-m (d) 844 N-m

04. A block brake shown below has a face width of 300 mm and a mean coefficient of friction of 0.25. For an activating force of 400 N, the braking torque in N-m is (GATE-ME-07)



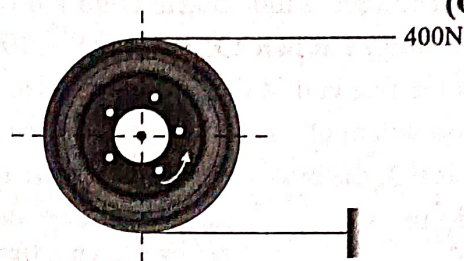
- (a) 30 (b) 40 (c) 45 (d) 60

05. A band brake having band width of 80 mm, drum diameter of 250 mm, coefficient of friction 0.25 and angle of wrap of 270 degrees is required to exert a friction torque of 1000 N-m. The maximum tension (in kN) developed in the band is (GATE-ME-10)

- (a) 1.88 (b) 3.56 (c) 6.12 (d) 11.56

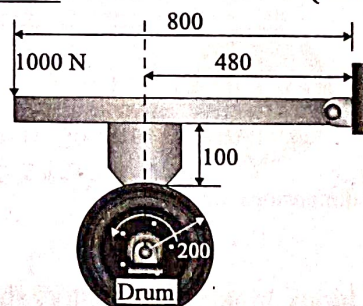
06. A force of 400 N is applied to the brake drum of 0.5 m diameter in a band – brake system as shown in the figure, where the wrapping angle is 180° . If

the coefficient of friction between the drum and the band is 0.25, the braking torque applied, in N.m is (GATE-ME-12)



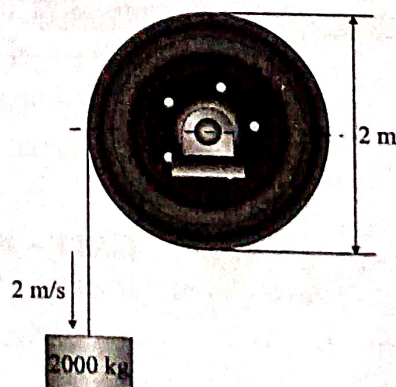
- (a) 100.6 (b) 54.4 (c) 22.1 (d) 15.7

07. A drum brake is shown in the figure. The drum is rotating in anticlockwise direction. The coefficient of friction between drum and shoe is 0.2. The dimensions shown in the figure are in mm. The braking torque (in N.m) for the brake shoe is (GATE-14-SET-3)

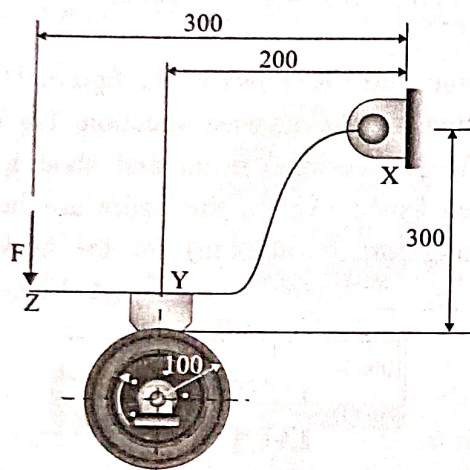


all dimensions are in mm

08. A mass of 2000 kg is currently being lowered at a velocity of 2 m/s from the drum as shown in the figure. The mass moment of inertia of the drum is 150 kg-m^2 . On applying the brake, the mass is brought to rest in a distance of 0.5 m. The energy absorbed by the brake (in kJ) is (GATE - 16 - SET - 2)

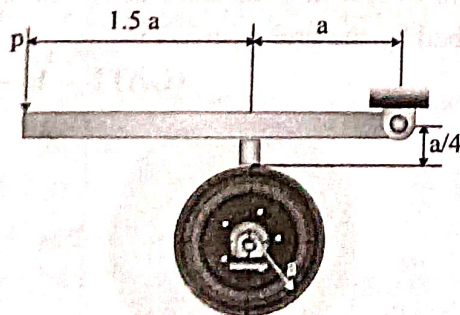


09. The schematic of an external drum rotating clockwise engaging with a short shoe is shown in the figure. The shoe is mounted at point Y on a rigid lever XYZ hinged at point X. A force $F = 100 \text{ N}$ is applied at the free end of the lever as shown. Given that the coefficient of friction between the shoe and the drum is 0.3, the braking torque (in Nm) applied on the drum is _____ (correct to two decimal places).
(GATE - 18 - SET - 1)



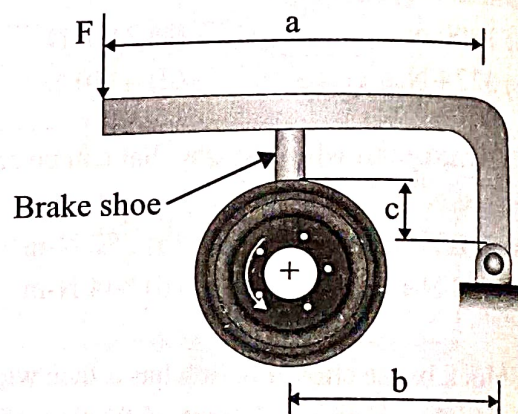
(All dimensions are in mm)

10. A single block brake with a short shoe and torque capacity of 250 N-m is shown. The cylindrical brake drum rotates anticlockwise at 100 rpm and the coefficient of friction is 0.25. The value of a , in mm (round off to one decimal place), such that the maximum actuating force P is 2000 N, is _____.

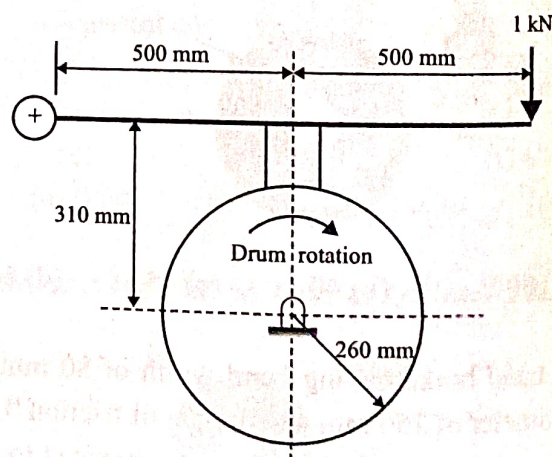


GATE - 19 - SET - 1)

11. A short shoe external drum brake is shown in the figure. The diameter of the brake drum is 500 mm. The dimensions $a = 1000 \text{ mm}$, $b = 500 \text{ mm}$ and $c = 200 \text{ mm}$. The coefficient of friction between the drum and the shoe is 0.35. The force applied on the lever $F = 100 \text{ N}$ as shown in the figure. The drum is rotating anti-clockwise. The braking torque on the drum is _____ N-m (round off to two decimal places).
(GATE - 19 - SET - 2)



12. A short shoe drum (radius 260 mm) brake is shown in the figure. A force of 1 kN is applied to the lever. The coefficient of friction is 0.4.



The magnitude of the torque applied by the brake is _____ N.m (round off to one decimal place).

(GATE-21_SET-1)

KEY & Detailed Solutions

ONE MARK QUESTIONS

Q1. 2.248	Q2. (a)	Q3. 4.11		
-----------	---------	----------	--	--

TWO MARKS QUESTIONS

Q1. (d)	Q2. (b)	Q3. (b)	Q4. (c)	Q5. (d)
Q6. (b)	Q7. 64 N-m	Q8. 14.11	Q9. 8.18	
Q10. 212.5	Q11. 20.35	Q12. 200		

One Mark Solutions

01. Ans: 2.248 (range 2.2 to 2.3)

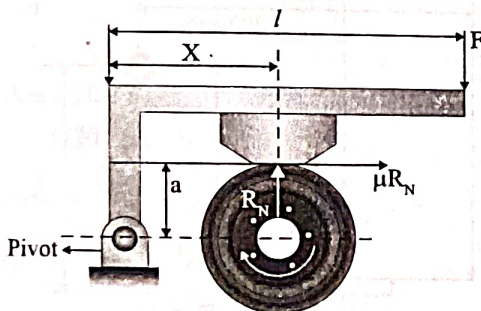
$$\text{Sol: } \frac{F_1}{F_2} = e^{\mu\theta} = e^{0.25 \times \frac{3\pi}{2}} = 3.248$$

$$F_1 = 3.248 \text{ N}$$

$$\text{Torque, } T = (F_1 - F_2) \times r = (3.248 - 1) \times 1 = 2.248 \text{ N-m}$$

02. Ans: (a)

Sol:



For clockwise rotation

$$\sum M_{\text{PIVOT}} = 0$$

$$F \times l - R_N \times x + \mu R_N \times a = 0$$

When applied moment and frictional moment are in same direction it is called self energizing brake.

03. Ans: 4.11

Sol: Ratio of tension,

$$\frac{T_1}{T_2} = e^{\mu\theta}$$

$$= e^{0.3 \times 270 \times \frac{\pi}{180}}$$

$$= 4.11$$

Two Marks Solutions

01. Ans: (d)

$$\text{Sol: } \frac{T_1}{T_2} = 3, \quad \theta = 180^\circ$$

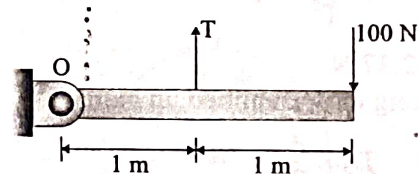
$$\frac{T_1}{T_2} = e^{\mu\theta}$$

$$\Rightarrow 3 = e^{\mu \times \frac{180 \times \pi}{180}}$$

$$\Rightarrow \mu = 0.35$$

02. Ans: (b)

Sol:



$$\theta = 270^\circ ; \quad T = 100 \text{ N,}$$

$$\mu = 0.5 ; \quad \sum M_0 = 0,$$

$$100 \times 2 = T \times 1$$

$$\Rightarrow T = 200 \text{ N}$$

$$\frac{T_1}{T_2} = e^{\mu\theta} = \frac{T_1}{200} = e^{0.5 \times 270 \times \frac{\pi}{180}}$$

(Taking $T = T_2$)

$$\Rightarrow T_1 = 2110 \text{ N}$$

03. Ans: (b)

$$\begin{aligned} \text{Sol: Torque, } T &= (T_1 - T_2) r \\ &= (2110 - 200) 0.2 \\ &= 382 \text{ N-m} \end{aligned}$$

04. Ans: (c)

$$\text{Sol: } \sum M_{\text{pivot}} = 0$$

$$P \times 600 = R_N \times 200$$

$$\Rightarrow 400 \times 600 = R_N \times 200$$

$$\Rightarrow R_N = 1200 \text{ N}$$

Friction force,

$$\mu R_N = 0.25 \times 1200 \text{ N} = 300 \text{ N}$$

$$\text{Torque, } T = 300 \times 0.150 = 45 \text{ N-m}$$

05. Ans: (d)

Sol: Torque, $T = (T_1 - T_2) r$

$$1000 \text{ N-m} = (T_1 - T_2) \times 0.125$$

$$\Rightarrow T_1 - T_2 = 8000 \text{ N} \dots\dots\dots(1)$$

$$\frac{T_1}{T_2} = e^{\mu \theta} \Rightarrow \frac{T_1}{T_2} = e^{0.25 \times 270 \times \frac{\pi}{180}} = 3.25 \dots\dots\dots(2)$$

$$\text{From (1) and (2), } T_2 = \frac{8000}{2.25} = 3555 \text{ N}$$

$$T_1 = 11555 \text{ N}$$

06. Ans: (b)

$$\text{Sol: } \frac{T_1}{T_2} = e^{\mu \theta} \Rightarrow \frac{400}{T_2} = e^{0.25 \times \frac{\pi}{180} \times 180} = 2.19$$

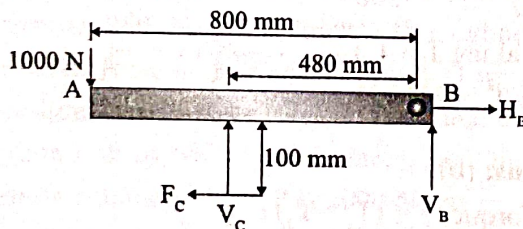
$$T_2 = 182.37 \text{ N}$$

$\therefore$ Braking torque applied, in N.m,

$$T_B = (T_1 - T_2) \times \frac{d}{2} \\ = (400 - 182.37) \times \frac{0.5}{2} = 54.4 \text{ N-m}$$

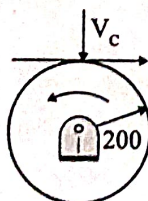
07. Ans: 64 N-m

Sol: FBD of shoe bar:



FBD of Drum Brake :

$$\Sigma M_B = 0$$



$$V_C \times 480 + F_C \times 100 - 1000 \times 800 = 0$$

$$F_C = \mu V_C = 0.2 V_C$$

$$480 V_C + 0.2 V_C \times 100 = 800000$$

$$500 V_C = 800000$$

$$V_C = 1600 \text{ N}$$

$$F_C = 0.2 V_C = 0.2 \times 1600 = 320 \text{ N}$$

$$M = 0.2 \times F_C = 0.2 \times 320 = 64 \text{ N-m}$$

08. Ans: 14.11 (range 14.1 to 14.3)

$$\text{Sol: } K.E_T = \frac{1}{2} m (V_1^2 - V_2^2)$$

$$= \frac{1}{2} \times 2000 \times (2^2 - 0) = 4000 \text{ J}$$

$$K.E_R = \frac{1}{2} I (\omega_1^2 - \omega_2^2)$$

$$= \frac{1}{2} \times 150 \times (2^2 - 0) = 300 \text{ J}$$

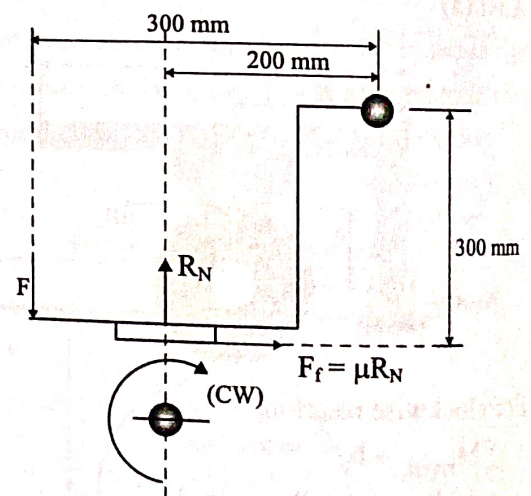
$$P.E = mgh = 2000 \times 9.81 \times 0.5 = 9810 \text{ J}$$

$$\text{Total, } E = 4000 + 300 + 9810$$

$$= 14110 \text{ J} = 14.11 \text{ kJ}$$

09. Ans: 8.18

Sol:



$$\therefore F \times 300 - R_N \times 200 + \mu R_N \times 300 = 0$$

$$\text{But } F = 100 \text{ N, } \mu = 0.3$$

$$100 \times 300 - R_N \times 200 + 0.3 \times R_N \times 300 = 0$$

$$\therefore R_N = 272.72 \text{ N}$$

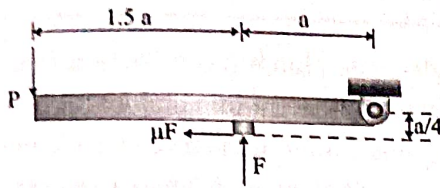
$$\therefore \text{Braking Torque} = \mu R_N \times \frac{D}{2}$$

$$= 0.3 \times 272.72 \times 0.1 \text{ N-m}$$

$$T_B = 8.18 \text{ N-m}$$

10. Ans: 212.5

Sol:



$$\mu = 0.25$$

$$\text{Torque} = 250 \text{ Nm}$$

$$\mu F \times a = 250$$

$$\Rightarrow 0.25 \times F \times a = 250$$

$$\Rightarrow F \times a = 1000 \text{ -----(i)}$$

In FBD of block and lever,

$$\sum M_{\text{fulcrum}} = 0$$

$$F \times a + \mu F \times \frac{a}{4} = P \times 2.5a$$

$$F \times a \left(1 + \frac{\mu}{4}\right) = P \times 2.5a$$

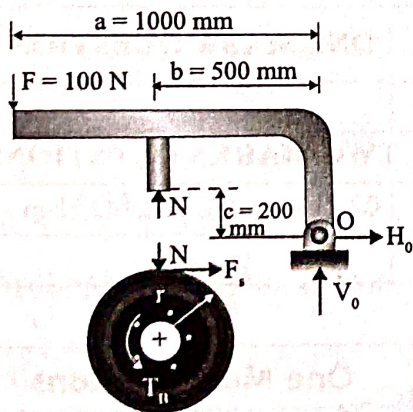
$$1000 \left(1 + \frac{0.25}{4}\right) = 2000 \times 2.5a$$

$$\Rightarrow a = 0.2125 \text{ m} = 212.5 \text{ mm}$$

11. Ans: 20.35

Sol: W.r.t FBD of link

$$F_s = \mu_s N = 0.35 \times N;$$



Taking summation of moment about O = zero

$$\sum M_O = 0$$

$$N \times 500 - F_s \times 200 - F \times 1000 = 0$$

$$N \times 500 - 0.35 \times N \times 200 - 100 \times 1000 = 0$$

$$N = \frac{100 \times 1000}{(500 - 70)} = 232.558 \text{ N}$$

With respect to FBD of drum

$$\text{Braking torque } T_B = F_s \times r$$

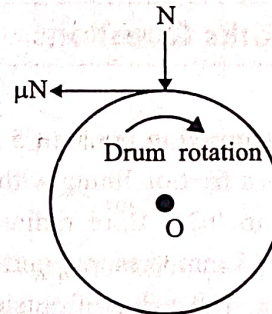
$$= 0.35 \times 232.558 \times \left(\frac{500}{2 \times 1000}\right)$$

$$T_B = 20.35 \text{ N-m}$$

12. Ans: (200)

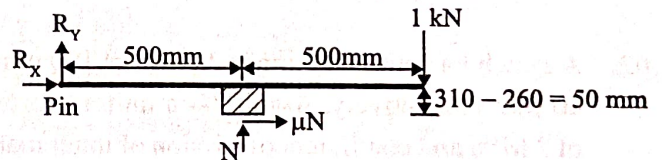
Sol:

(i) FBD of drum :



$$\text{Braking torque} = \mu N R \text{ ---- (i)}$$

(ii) FBD of Lever :



$$\sum M_{\text{pin}} = 0$$

$$N \times 500 + 0.4 N \times 50 = 10^3 \times 1000$$

$$\therefore N = 1923 \text{ N}$$

From equation (i)

$$\text{Braking torque} = \mu N R$$

$$= 0.4 \times 1923 \times 260 = 200 \text{ Nm}$$

Chapter 8

Clutches

One Mark Question

01. Axial operation claw clutches having self- locking tooth profile (GATE-ME-87)
- can be disengaged at any speed
 - can be disengaged only when unloaded
 - can be engaged only when unloaded
 - can work only with load

Two Marks Questions

01. A disk clutch is required to transmit 5 kW at 2000 rpm. The disk has a friction lining with Coefficient of friction equal to 0.25. Bore radius of friction lining is equal to 25 mm. Assume uniform contact pressure of 1 MPa. The value of outside radius of the friction lining is (GATE-ME-06)
- 39.4 mm
 - 49.5 mm
 - 97.9 mm
 - 142.9 mm
02. A clutch has outer and inner diameters 100 mm and 40 mm respectively. Assuming a uniform pressure of 2 MPa and coefficient of friction of inner material 0.4, the torque carrying capacity of the clutch is (GATE-ME-08)
- 148 Nm
 - 196 Nm
 - 372 Nm
 - 490 Nm
03. A disc clutch with a single friction surface has coefficient of friction equal to 0.3. The maximum pressure which can be imposed on the friction material is 1.5 MPa. The outer diameter of the clutch plate is 200 mm and its internal diameter is 100 mm. Assuming uniform wear theory for the clutch plate, the maximum torque (in N.m) that can be transmitted is _____ (GATE-14-SET-2)

04. A single-plate clutch has a friction disc with inner and outer radii of 20 mm and 40 mm, respectively. The friction lining in the disc is made in such a way that the coefficient of friction μ varies radially as $\mu = 0.01 r$, where r is in mm. The clutch needs to transmit a friction torque of 18.85 kN.mm. As per uniform pressure theory, the pressure (in MPa) on the disc is _____ (GATE - 17 - SET - 2)
05. In a disc-type axial clutch, the frictional contact takes place within an annular region with outer and inner diameters 250 mm and 50 mm, respectively. An axial force F_1 is needed to transmit a torque by a new clutch. However, to transmit the same torque, one needs an axial force F_2 when the clutch wears out. If contact pressure remains uniform during operation of a new clutch while the wear is assumed to be uniform for an old clutch, and the coefficient of friction does not change, then the ratio F_1/F_2 is _____ (round off to 2 decimal places). (GATE-20-SET-1)

KEY & Detailed Solutions

ONE MARK QUESTION

01. (a)				
---------	--	--	--	--

TWO MARKS QUESTIONS

01. (a)	02. (b)	03. 530.142 N-m	04. 0.5
05. 0.87			

One Mark Solutions

01. Ans: (a)

Sol: Axial operation claw clutches having self- locking tooth profile can be disengaged at any speed.

Two Marks Solutions

01. Ans: (a)

Sol: $T = \frac{60p}{2\pi N} = \frac{60 \times 5 \times 10^3}{2\pi \times 2000} = 23.87 \text{ N-m.}$

$W = P \times \pi(r_1^2 - r_2^2)$

$T = \frac{2}{3} \mu W \left[\frac{r_1^3 - r_2^3}{r_1^2 - r_2^2} \right] = \frac{2}{3} \mu P \pi (r_1^3 - r_2^3)$

$23.87 = \frac{2}{3} \times 0.25 \times 1 \times \pi (r_1^3 - 25^3)$

$\therefore r_1 = 39.41 \text{ mm}$

02. Ans: (b)

Sol: $T = \frac{2}{3} \mu W \left(\frac{r_1^3 - r_2^3}{r_1^2 - r_2^2} \right)$
 $= \frac{2}{3} \times 0.4 \times 2 \times \pi (r_1^2 - r_2^2) \left(\frac{r_1^3 - r_2^3}{r_1^2 - r_2^2} \right)$
 $= \frac{2}{3} \times 0.4 \times 2 \times \pi \times (50^3 - 20^3) = 196 \text{ N-m}$

03. Ans: 530.142 N-m

Sol: $\mu = 0.3, P_{\max} = 1.5 \text{ MPa}$

$r_1 = \frac{200}{2} = 100 \text{ mm}, r_2 = \frac{100}{2} = 50 \text{ mm.}$

$W = 2\pi c(r_1 - r_2) = 2\pi \times P_{\max} \times r_2(r_1 - r_2)$
 $= 2\pi \times 1.5 \times 50 (100 - 50)$
 $= 2\pi \times 1.5 \times 50 \times 50 = 23561.9 \text{ N}$

$T_w = \mu W \times \frac{(r_1 + r_2)}{2} = 0.3 \times 23561.9 \times \frac{(100 + 50)}{2}$
 $= 530142.75 \text{ N-mm} = 530.142 \text{ N-m}$

04. Ans: 0.5

Sol: $r_1 = 40 \text{ mm}, r_2 = 20 \text{ mm}, \mu = 0.01$

$T = 18.85 \text{ kN-mm}, p = c$

Torque, $\delta T = \mu P \times 2\pi r dr \times r$

$T = P \times 2\pi \int_{r_2}^{r_1} \mu r^2 dr$

$= P \times 2\pi \times 0.01 \times \int_{r_2}^{r_1} r^3 dr$

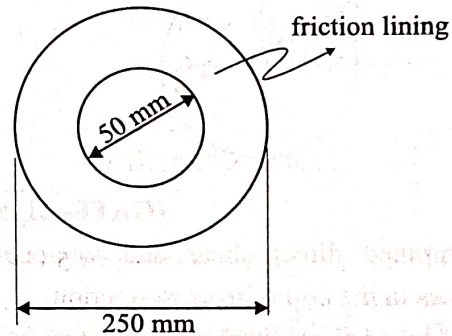
$= P \times 2\pi \times 0.01 \times \left[\frac{r_1^4 - r_2^4}{4} \right]$

$18.85 \times 10^3 = P \times 2\pi \times 0.01 \times \left[\frac{40^4 - 20^4}{4} \right]$

$\Rightarrow P = 0.5 \text{ N/mm}^2$

05. Ans: 0.87

Sol:



Uniform pressure theory:

Torque transmission, $T_p = \mu \omega_p \cdot r_p$

$= \mu \cdot F_1 \cdot \frac{2}{3} \cdot \frac{r_o^3 - r_i^3}{r_o^2 - r_i^2} = \mu \cdot F_1 \cdot \frac{2}{3} \cdot \left(\frac{125^3 - 25^3}{125^2 - 25^2} \right)$

$T_p = \frac{1550}{18} \mu \cdot F_1$ -----(1)

Uniform Wear Theory :

Torque transmission, $T_w = \mu \cdot W_w \cdot r_w$

$= \mu \cdot F_2 \cdot \frac{r_o + r_i}{2}$

$T_w = \mu \cdot F_2 \cdot \frac{125 + 25}{2}$

$T_w = 75 \mu \cdot F_2$ -----(2)

from equation 1 & 2 $\rightarrow$

$\frac{T_p}{T_w} = \frac{\frac{1550}{18} \cdot \mu \cdot F_1}{75 \cdot \mu \cdot F_2}$

Here, $T_p = T_w$

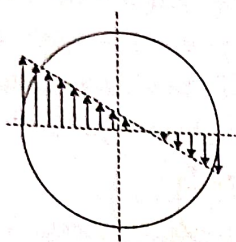
$\frac{F_1}{F_2} = \frac{75 \times 18}{1550}$

$\frac{F_1}{F_2} = 0.871$

Chapter 9 Springs & Belt Pulley

One Mark Question

01. Shear stress distribution on the cross-section of the coil wire in a helical compression spring is shown in the figure. This shear stress distribution represents



(GATE-21_SET-1)

- combined direct shear and torsional shear stress in the coil wire cross-section
 - combined direct shear and torsional shear stress along with the effect of stress concentration at inside edge of the coil wire cross-section
 - direct shear stress in the coil wire cross-section
 - torsional shear stress in the coil wire cross-section
02. A 150 mm wide polyamide flat belt is transmitting 15 kW power through a belt-pulley system. The driving pulley of 150 mm pitch diameter is rotating at 200 RPM. If F_1 is the belt tension on high tension side, and F_2 is the belt tension on low tension side, then the difference in belt tensions $\Delta F = F_1 - F_2$ in N is _____. (round off to one decimal place)

(GATE_PI-21)

One Mark Solutions

01. Ans: (a)

Sol: The figure represents variation of shear stress considering direct shear loading and torsional loading.

As the variation is linear it is observed that the effect of curvature is not considered.

02. Ans: (9549.3)

Sol: Given, $P = 15 \text{ kW}$,
 $d = 150 \text{ mm} = 0.15 \text{ m}$
 $N = 200 \text{ rpm}$

For belt drive,

$$P = \frac{2\pi N}{60} \times \{F_1 - F_2\} \times \frac{d}{2}$$

$$\therefore 15 \times 10^3 = \frac{2\pi \times 200}{60} \times \Delta F \times \frac{0.15}{2}$$

$$\therefore \Delta F = 9549.3 \text{ N}$$

KEY & Detailed Solutions

ONE MARK QUESTION

01. (a)

02. 9549.3